

Chapter 1 The Nature of Science

Science is a systematic enterprise that builds knowledge in the form of testable explanations about the universe. It relies on observation and evidence.

Chapter 2 Motion in One Dimension

Position, velocity and acceleration describe motion. The equations: $v = u + a t$, $s = u t + 0.5 a t^2$

Chapter 3 Newton Laws of Motion

Newton first law: inertia. Newton second law: $F = m a$. Newton third law: every action has an equal and opposite reaction.

Chapter 4 Work Energy and Power

Work done $W = F d \cos \theta$. Kinetic energy $K = 0.5 m v^2$. Power $P = W / t$.

Chapter 5 Optics

Light reflects and refracts. Snell law and the thin lens equation $1/f = 1/u + 1/v$ define image formation.